AALBORG UNIVERSITY

Introduction and Link Prediction Based on Structural Similarity

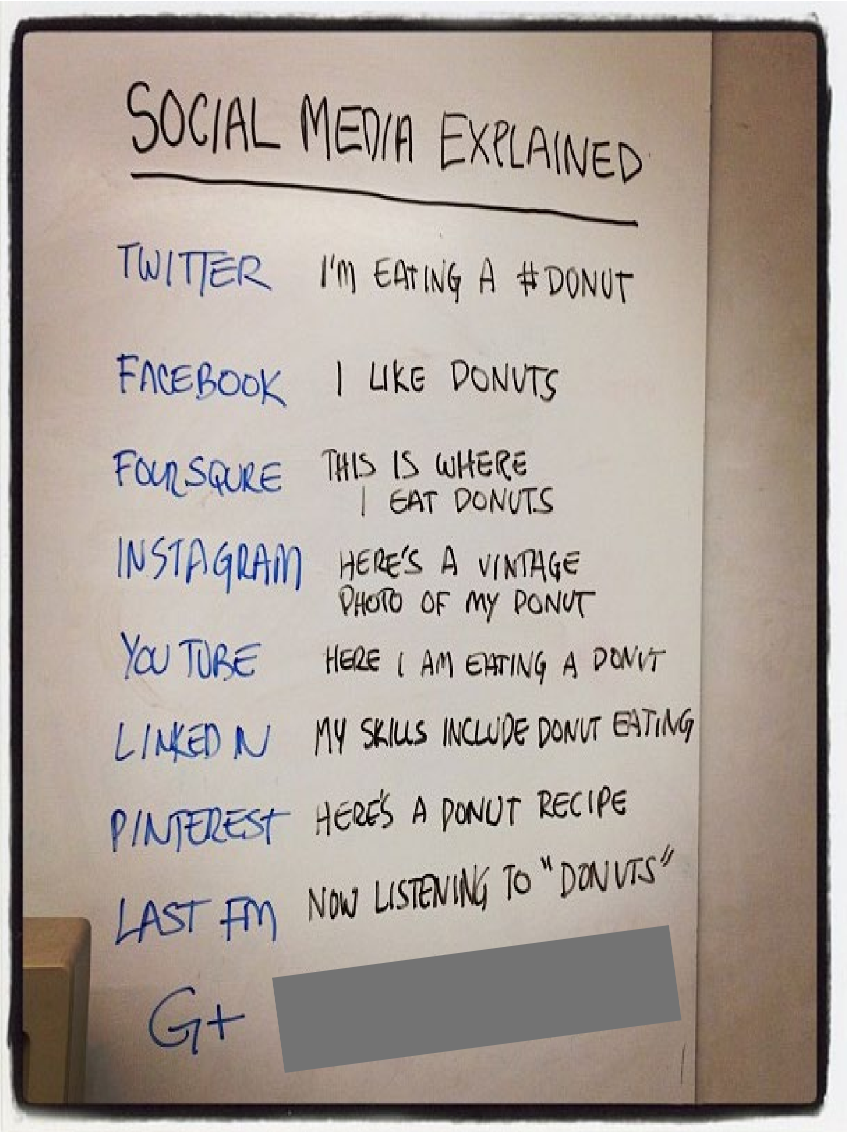
Peter Dolog
dolog@cs.aau.dk
http://people.cs.aau.dk/~dolog

Some slides from Bo Thiesson and Manfred Jaeger

Outline

- Introduction to social media analysis
- Basic graph theory
- Structural similarity
  - Local Measures
  - SimRank

Introduction



Douglas Wray - http://instagr.am/p/nm695/ @ThreeShipsMedia

Do people really talk about donuts? (on Twitter)

1 week of tweets mentioning “donut” or “doughnuts”

- Week of Feb 6-12, 2012.
- Matched ~180k messages

Can be used to find things such as

- In which locations do people eat donuts
- Preferred restaurants for eating donuts
- Preferred kind of donuts
- What people drink with donuts
- What is the mood, when eating donuts
- Etc.

Beyond donuts and social networks…

Drugs, diseases, and contagions

- Paul and Dredze 2011; Sadilek, Kautz and Silenzio 2012, **Denecke et al. 2013**

Crises, disasters, and wars

- Starbird et al. 2010; Al-Ani, Mark & Semaan 2010; Monroy-Hernandez et al. 2012

Public Sentiment

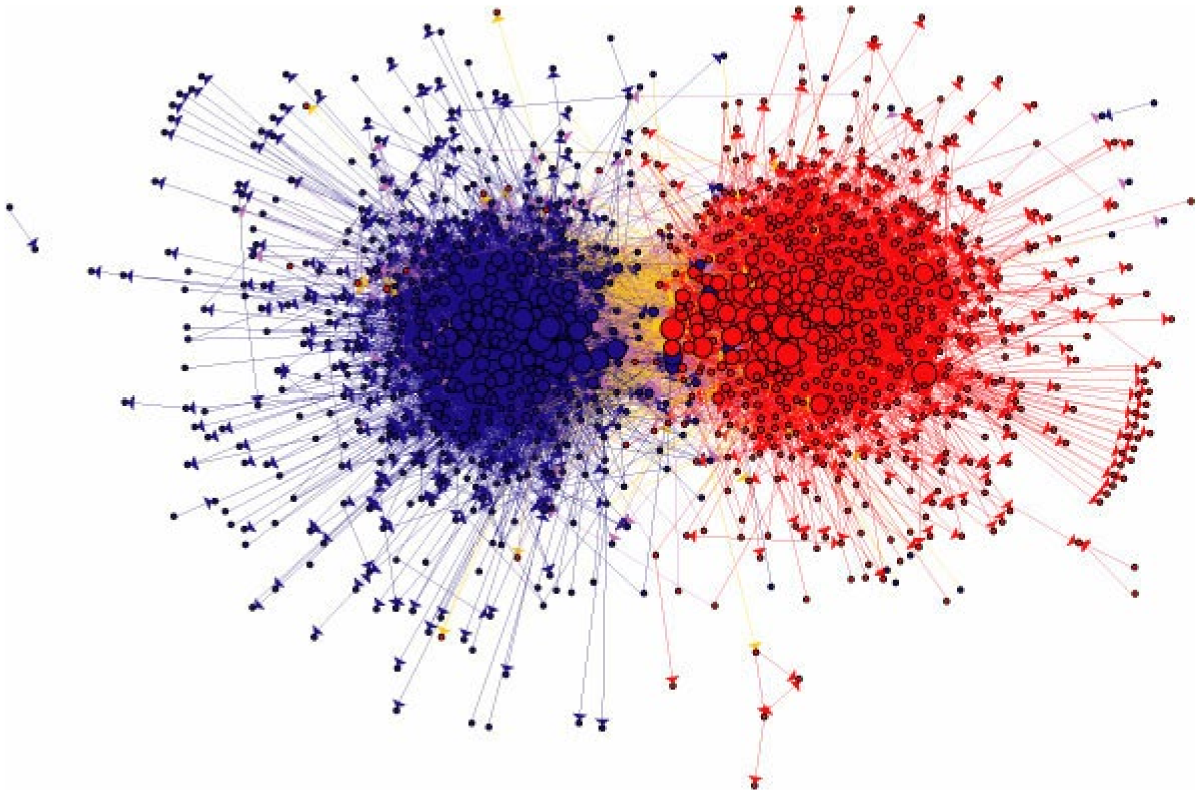
- Political and election indices, market insights

Molecule Interaction

Sensor Networks

Everyday life

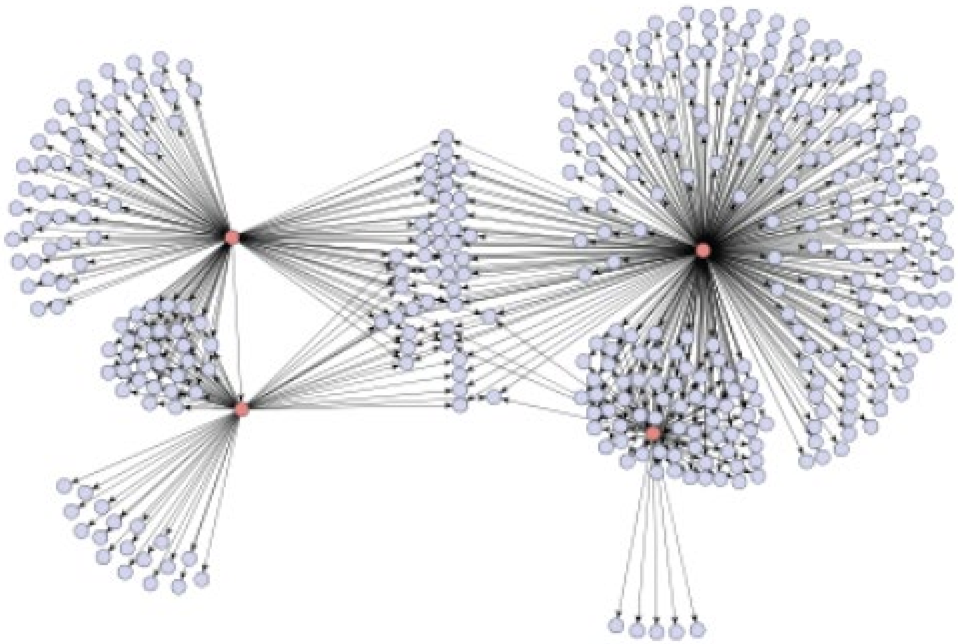
Political blogs

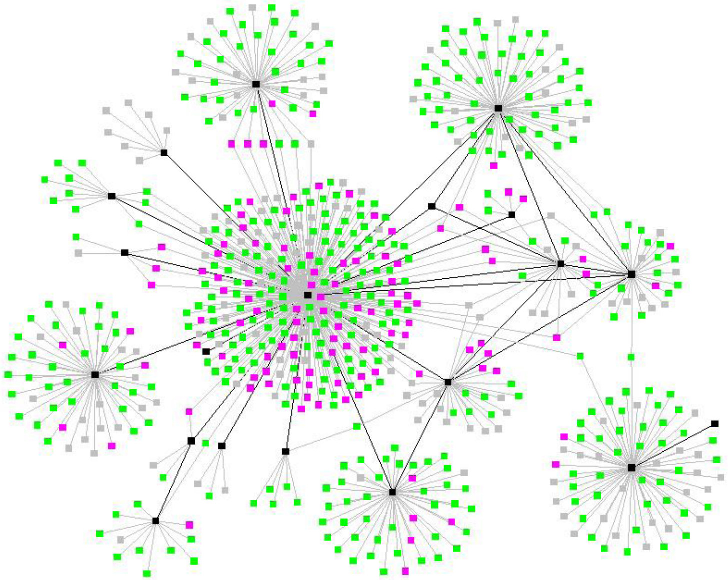


Adamic and Glance, 2005: The political blogosphere and the 2004 U.S. election

Social contagion cascade (viral marketing, Japanese graphic novel)

Biological contagion (tuberculosis outbreak)

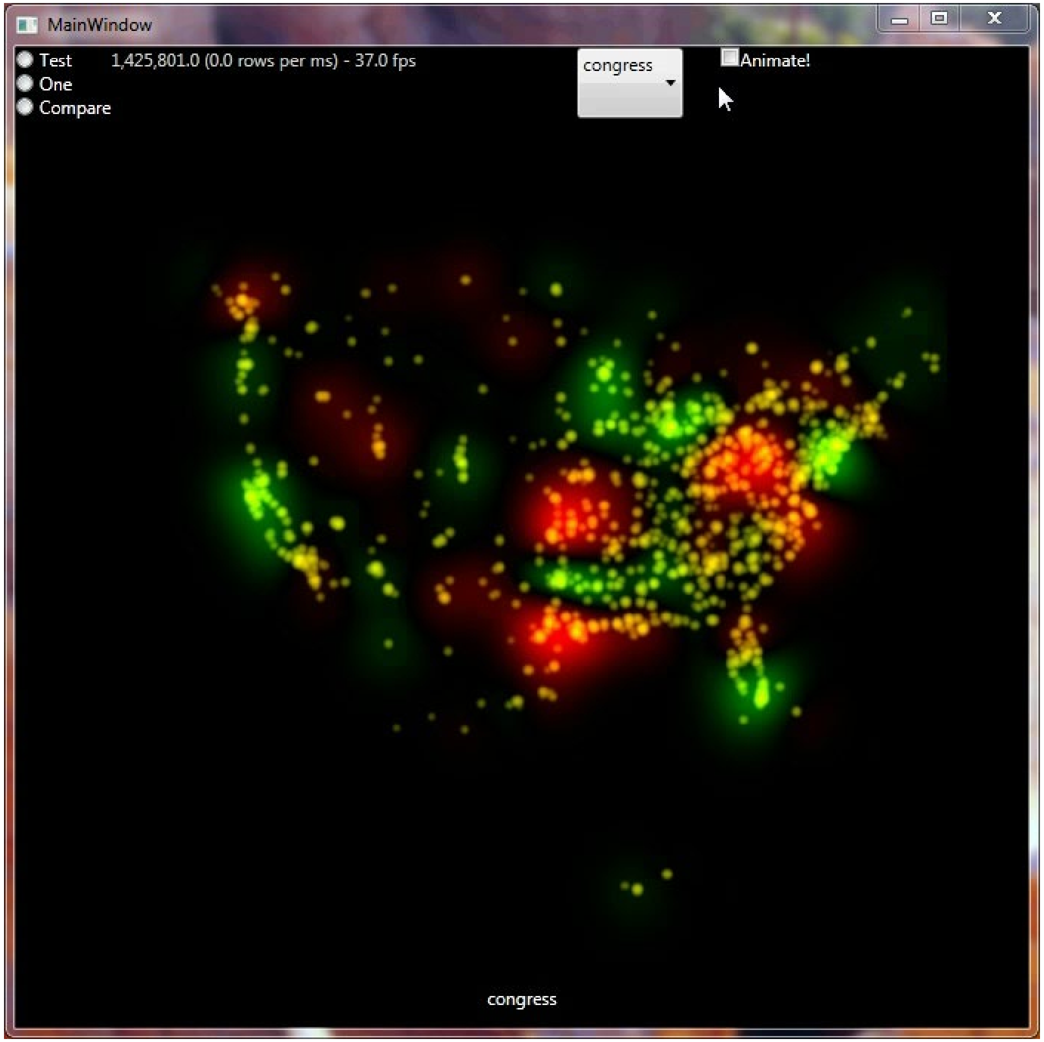




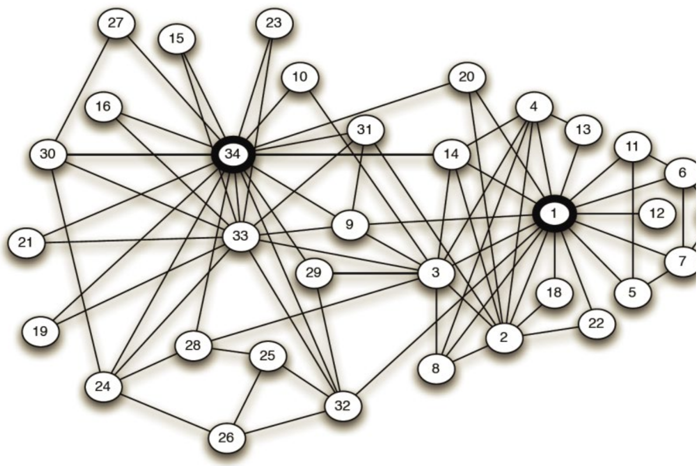
Leskovec, Adamic & Huberman, 2007. The dynamics of viral marketing.

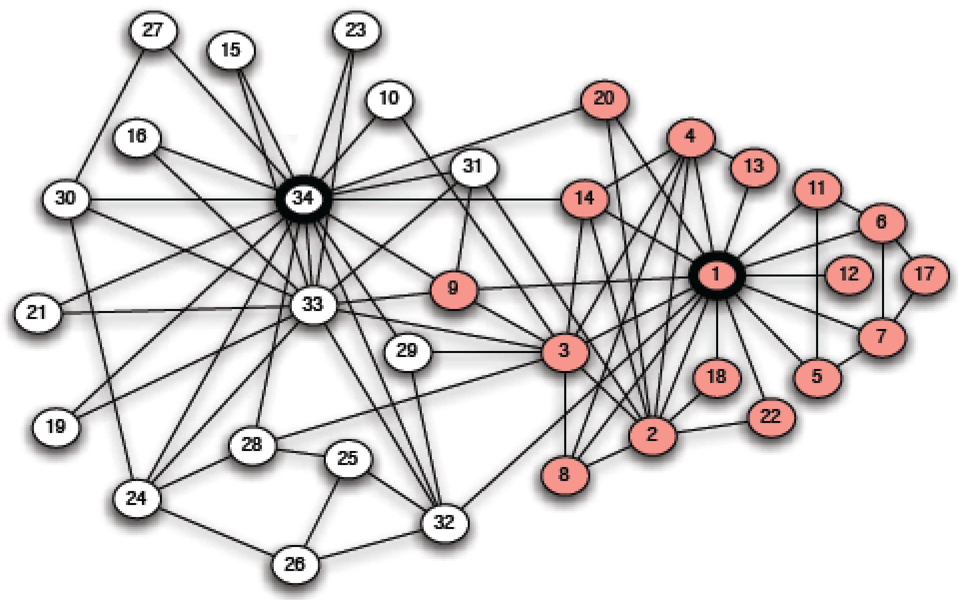
Andre *et al.*, 2007. Transmission network analysis to complement routine tuberculosis contact investigations.

Location based sentiment towards US Congress



Karate club friendships





Zachary, 1977. An information flow model for conflict and fission in small groups

Social Networks & Graph Theory

A social network is made up of actors (people, organizations, groups,…) and ties (social relationships, similarity,…)

A graph consists of nodes and edges between nodes.

$\Rightarrow$ Social network problems can in many cases be represented as a standard graph theoretic problems

Typical Tasks (some examples)

Node Classification

- Is a user in a social network going to vote democrat or republican?
- Is a sensor in a sensor network going to fail within the next 30 days?
- Has a computer in a computer network been hacked?
- Is an account a real person or a robot?

Link Prediction

- Are two proteins interacting?
- Is there a 'capital of' relation between Sacramento and California?
- Is user A going to become a follower of user B?

Network/Graph Classification

- Is a molecule a mutagen?

Link Prediction Interpretation

Given the **similarity of nodes**, can I predict where are the links between nodes?

- with whom I can be a friend
- whom I can follow
- whom I can recommend a product or information
- …

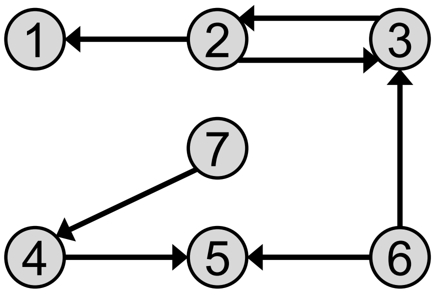
We can define similarity based on nodes properties or based on graph structure

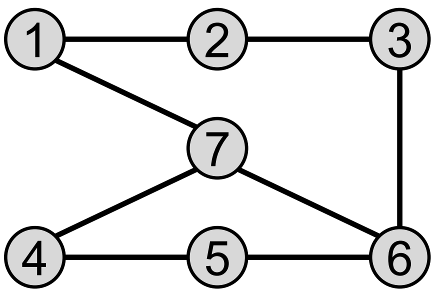
Today we are going to focus on graph structure

If two nodes are similar, there should be a link between them

Basic Graph Theory

Directed and Undirected graphs





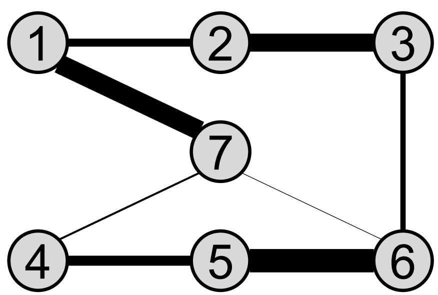
Graph: $G = (V, E)$

Nodes: $V = \{v_1, v_2, \dots, v_n\}$
Edges: $E = \{e_1, e_2, \dots, e_m\}$

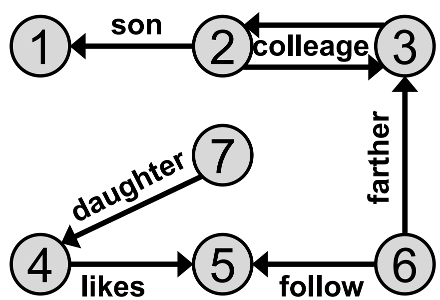
Directed graph
Well suited to represent follow/follower relationships seen on e.g. Twitter

Undirected graph
Well suited to represent undirectional relationships such as Facebook friends

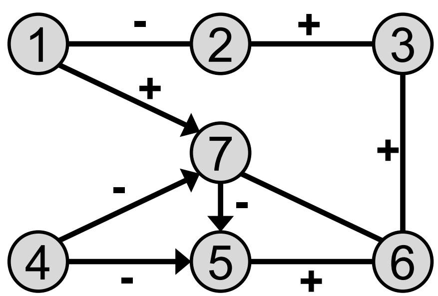
Weighted, Labeled, and Signed Graphs



Weighted (undirected) graph
Well suited to represent **intensity** of relationships such as number of interactions (e.g. email) or **similarity** (e.g. sentiments towards products)



Labeled (directed) graph
Well suited to represent **types** of relationships



Signed (mixed) graph
Well suited to represent **friends and foes** (undirected) or **social status** (directed)

Graph: $G = (V, E, W)$

Nodes: $V = \{v_1, v_2, \dots, v_n\}$
Edges: $E = \{e_1, e_2, \dots, e_m\}$
Edge weights/labels/signs: $W = \{w_1, w_2, \dots, w_m\}$

Possible to annotate nodes as well!

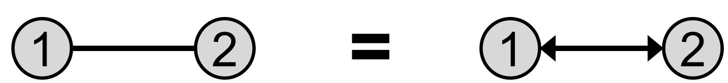
Degree (of a node)

For directed graph

- In-degree of node v_i = number of edges pointing into the node. d_i^{in}
- Out-degree of node v_i = number of edges pointing away from the node. d_i^{out}
- Degree = In-degree + Out-degree. d_i

For undirected graph

- Undirected edge = two opposite directed edges (aka. bi-directional or reciprocal edge)



- In-degree = Out-degree = number of edges connected to node
- Degree = 2 times the number of edges

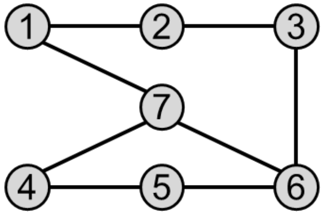
Degree Distribution

Degree distribution:

$$p_d = \frac{n_d}{n}$$

n_d :number of nodes with degree d
n :number of nodes

(Unrealistic) example:



$p_2 = 0/7, p_4 = 5/7, p_6 = 2/7$

Degree Distribution (for network)

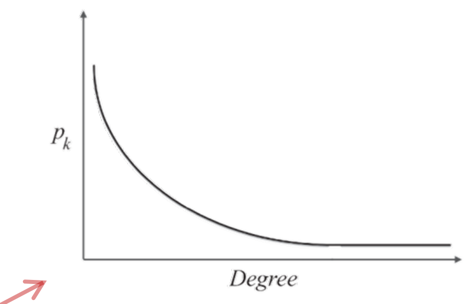
More realistic examples

- Many internet sites are visited <1,000 times a month whereas few are visited more than a million times daily.
- Social media users are often active on a few sites whereas a few individuals are active on hundreds of sites.
- On Facebook there exist many individuals with a few friends and a handful of users with many thousands of friends
- On Twitter many individuals follow (or are followed by) a few other individuals, and a few follows (or are followed) by a huge crowd

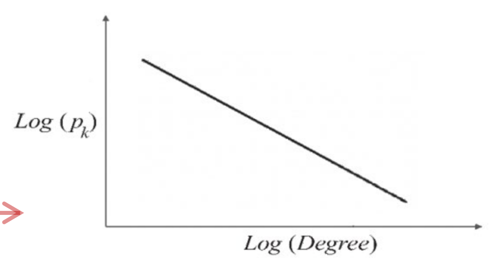
Me and you(?) ⟺ **Obama or Justin Bieber**

Power-Law Degree Distribution

- Many real-world (social) networks exhibit a *power-law* distribution.
- Power laws seem to dominate in cases where the quantity being measured can be viewed as a type of *popularity*. (e.g., node degree)
- A power-law distribution implies that small occurrences are common, whereas large instances are extremely rare.



(a) Power-Law Degree Distribution



(b) Log-Log Plot of Power-Law Degree Distribution

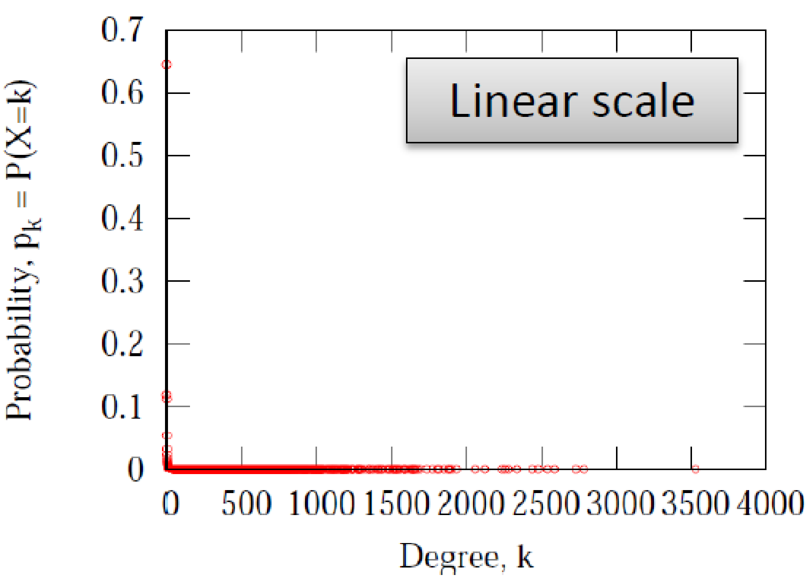
Power-Law degree distribution:

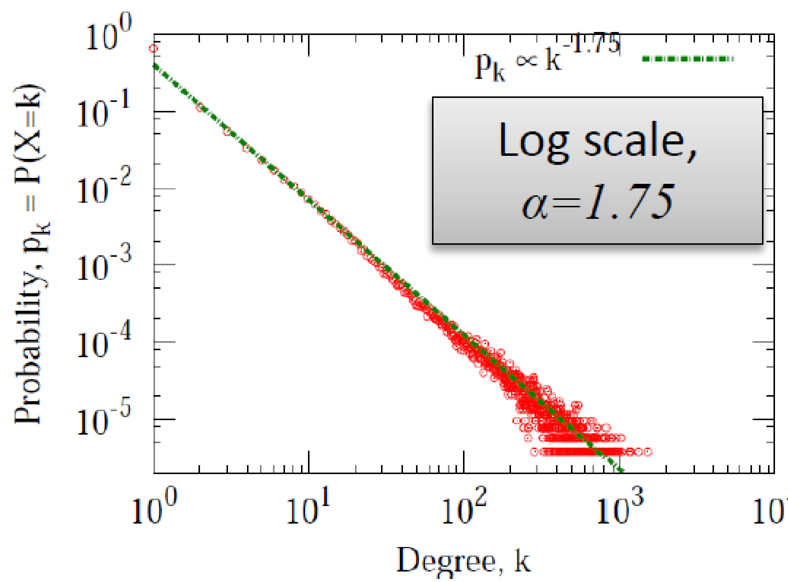
$$p_d = \beta d^{-\alpha}$$

$$\log(p_d) = \log(\beta) - \alpha \log(d)$$

α: the power-law exponent and its value is typically in the range of [2, 3]
β: power-law intercept

Flickr: Power-Law Degree Distribution





From: Jure Leskovec, ICML '09 tutorial

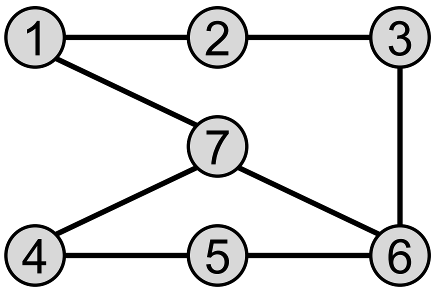
Graph representation

Visual graph good for intuition, but not for computation!

For computations we instead use

- Adjacency Matrix
- Adjacency List
- Edge list

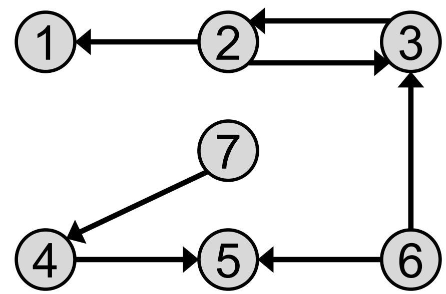
Adjacency Matrix



Properties

Symmetric

Asymmetric



A	1	2	3	4	5	6	7
1	0	1	0	0	0	0	1
2	1	0	1	0	0	0	0
3	0	1	0	0	0	1	0
4	0	0	0	0	1	0	1
5	0	0	0	1	0	1	0
6	0	0	1	0	1	0	1
7	1	0	0	1	0	1	0

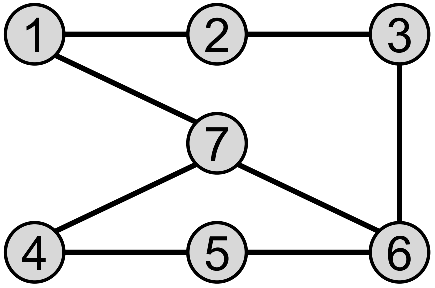
Weights/Labels/Signs instead of 0/1

A	1	2	3	4	5	6	7
1	0	0	0	0	0	0	0
2	1	0	1	0	0	0	0
3	0	1	0	0	0	0	0
4	0	0	0	0	1	0	0
5	0	0	0	0	0	0	0
6	0	0	1	0	1	0	0
7	0	0	0	1	0	0	0

Social media networks have very sparse adjacency matrices

Adjacency List (aka. sparse representation)

- Every node maintains a list of all the nodes that it connects to (in direction of arrow)
- Recall: undirected edges are bi-directional
- The list is usually sorted based on the node order or other preferences

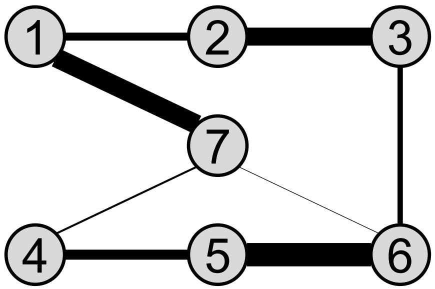


Node	Connects to
1	2, 7
2	1, 3
3	2, 6
4	5, 7
5	4, 6
6	3, 5, 7
7	1, 4, 6

Space efficient for sparse network

Adjacency List

- …and with Weights/Labels/Signs



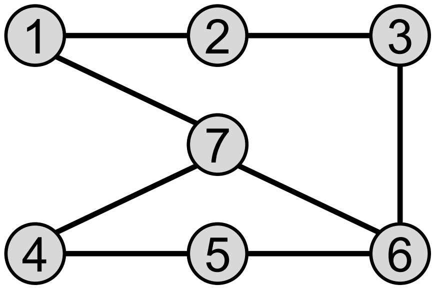
Node	Connects to
1	2:4, 7:11
2	1:4, 3:8
3	2:8, 6:2
4	5:6, 7:1
5	4:6, 6:14
6	3:2, 5:16, 7:1
7	1:11, 4:1, 6:1

Q: Does it look like something you have seen before?

A: Inverted index with terms replaced by nodes and documents replaced by “connects to” nodes

Edge List

- Each element is an edge and is usually represented as *(u, v)*, denoting that node *u* is connects to node *v* via a directed or undirected edge (semantic must be specified)



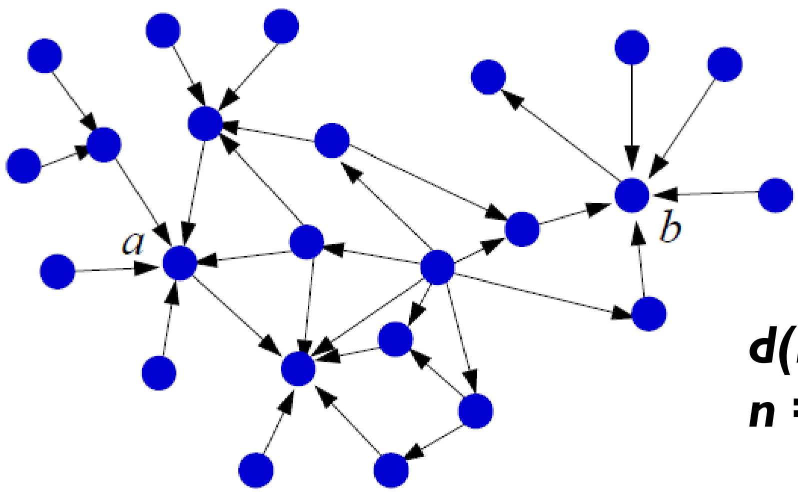
Edge list

(1,2)
(1,7)
(2,3)
(3,6)
(4,5)
(4,7)
(5,6)
(6,7)

- Not very useful for our tasks (personal opinion)

Structural Similarity Measures
(between nodes)

A simple **local measure**



d(i) **= in-degree (# incoming links)**
n **= # nodes**

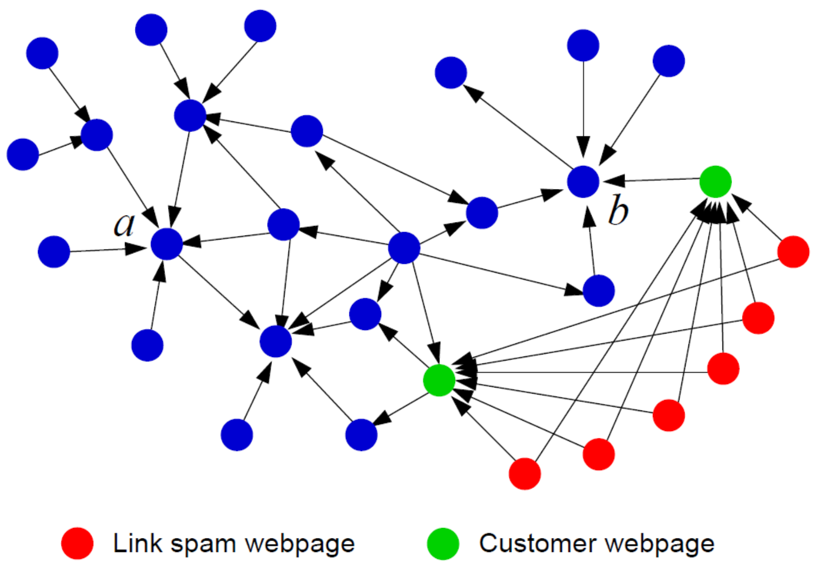
Prestige

P(i) = d(i) / (n-1)

Example: P(a) = P(b) = 5/23

Prestige (cont.)

Local measures can be easily manipulated by link spamming:

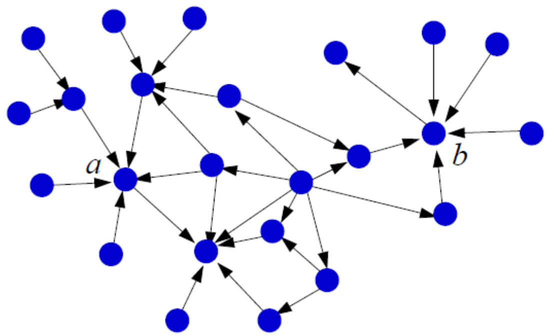


On the other hand, more difficult to manipulate global measures (though still possible)

A global prestige measure

Rank prestige:

$$P(j) = \sum_{i:i \to j} P(i)$$



Recursion: Prestige of page depends on

- Its in-degree, and
- The prestige of pages linking to it
- This does not directly define prestige – only a mutual relationship between values

Some common properties of those measures

Local Measures

- Easy to compute
- Easy to spam

Coverage

- Since they are local, they may miss data to be able to compute similarities for all pairs

Similarity Based on Node Degrees

Can be both, local and global

Already local measures perform well, but as said can be easily spammed

Structural Equivalence (Local Measures): Definitions

Vertex similarity:

$$Sim_{Vertex}(v_i, v_j) = |N(v_i) \cap N(v_j)|$$

Jaccard similarity

$$Sim_{Jaccard}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$$

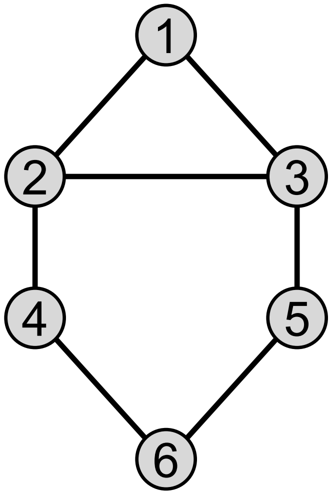
Cosine similarity

$$Sim_{Cosine}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)||N(v_j)|}}$$

Structural Equivalence: Example

$$Sim_{Vertex}(v_2, v_5) = |\{1,3,4\} \cap \{3,6\}| = 1$$

$$Sim_{Jaccard}(v_2, v_5) = \frac{|\{1,3,4\} \cap \{3,6\}|}{|\{1,3,4,6\}|} = \frac{1}{4}$$



$$Sim_{Cosine}(v_2, v_5) = \frac{|\{1,3,4\} \cap \{3,6\}|}{\sqrt{|\{1,3,4\}||\{3,6\}|}} = \frac{1}{\sqrt{6}}$$

$$Sim_{Pearson}(v_2, v_5) = \frac{1}{\sqrt{6}}$$

Similarities are NOT comparable across different similarity measures
Similarities are only comparable if computed with same measure

SIMILARITY BASED ON GLOBAL PRESTIGE OF THE NODE (RANDOM WALK)

Definition of PageRank
(Brin and Page '98)

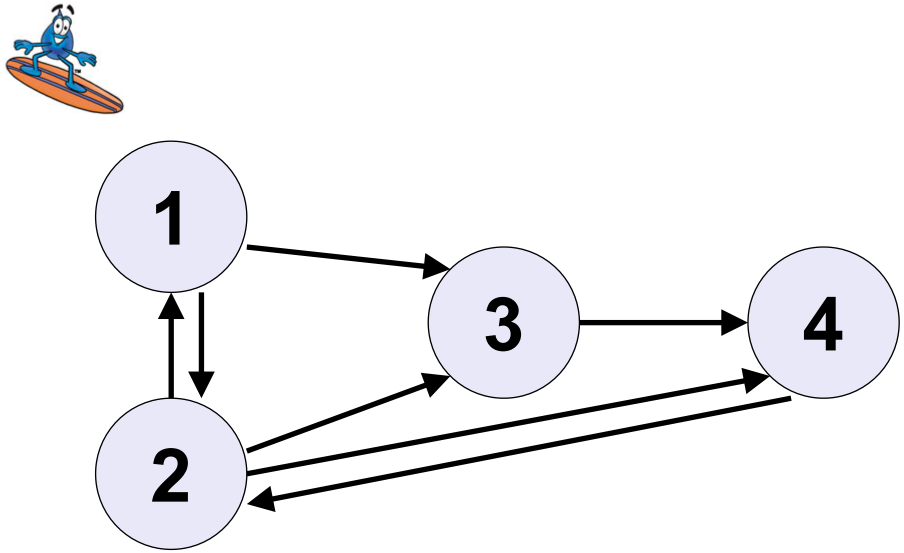
Consider the following infinite random walk (surf):

- Initially the surfer is at a random page
- At each step, the surfer proceeds
  - to a randomly chosen web page with probability α
  - to a randomly chosen successor of the current page with probability $1 - \alpha$

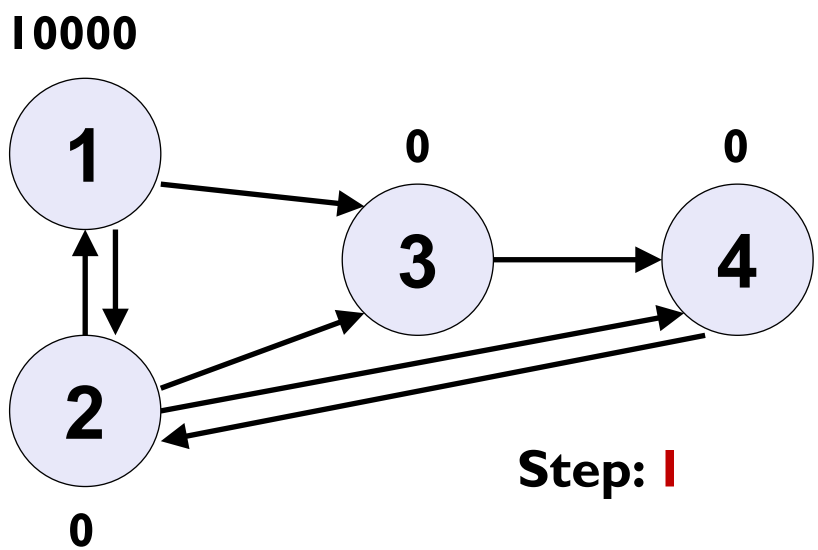
The PageRank of a page p is the fraction of steps the surfer spends at p in the limit. (A score between 0 and 1)

Notice: Using link structure only!

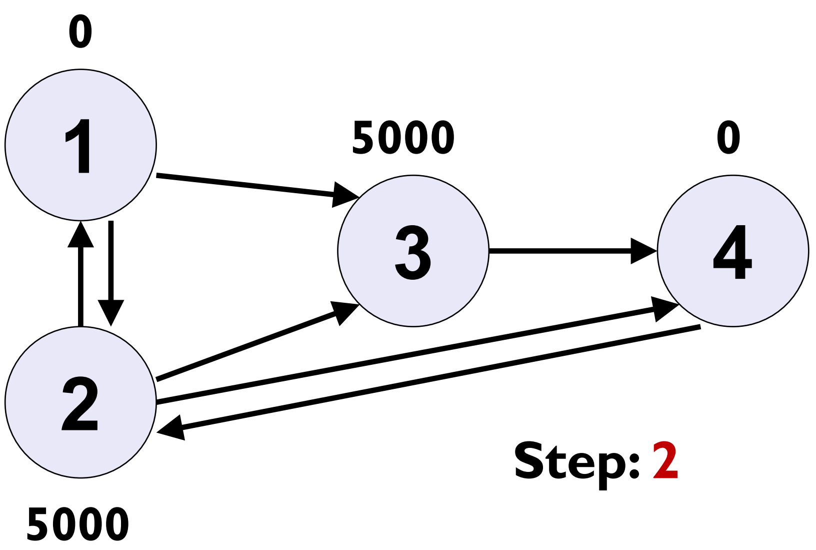
The Random Web Surfer



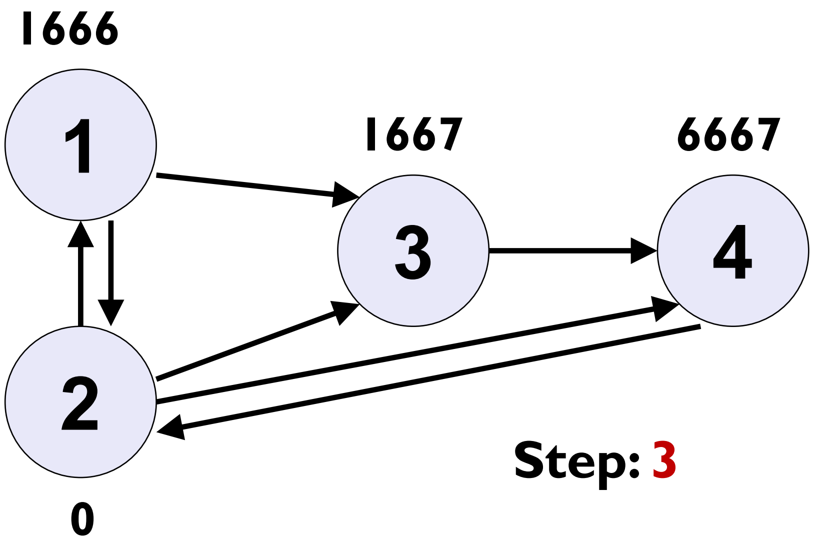
10.000 Random Surfers



10.000 Random Surfers



10.000 Random Surfers



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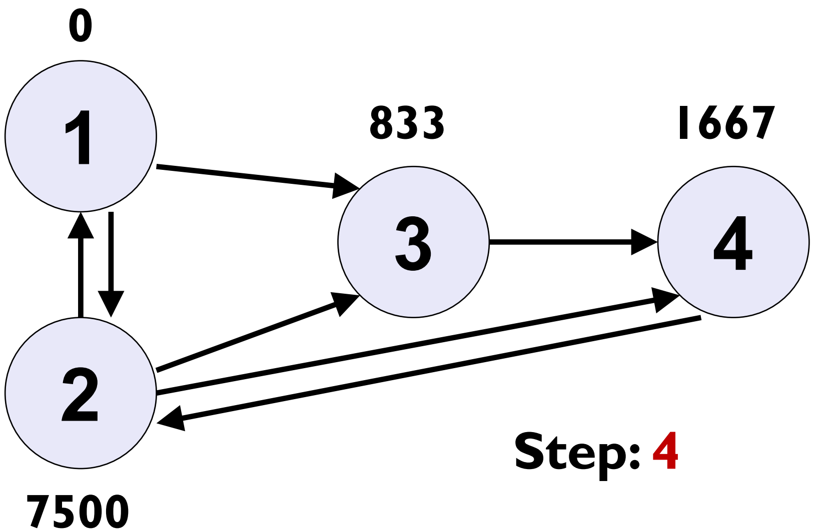
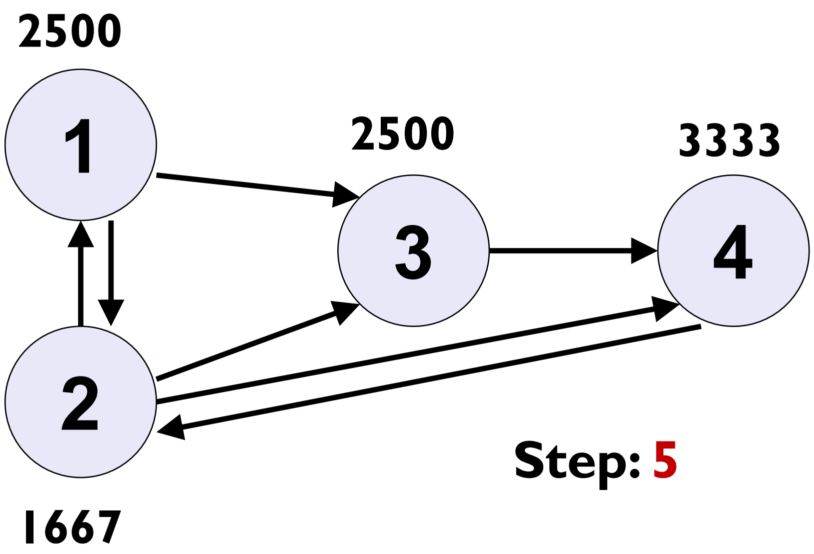


10.000 Random Surfers



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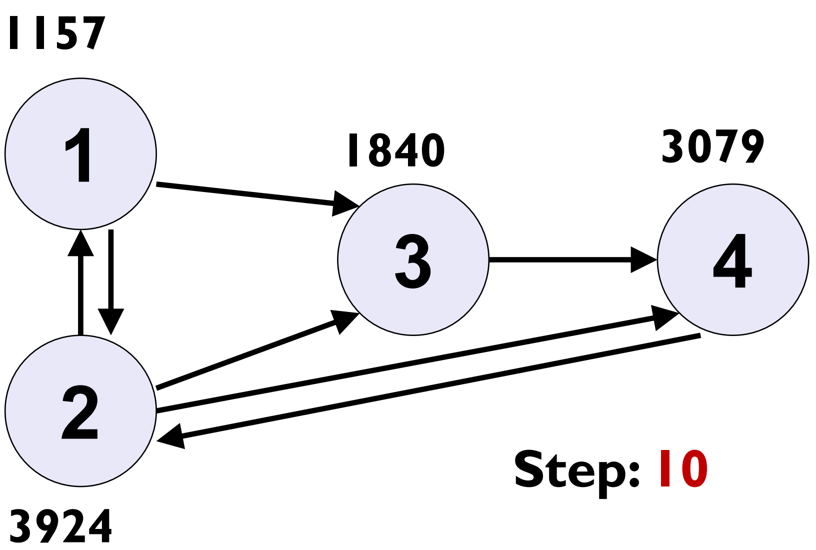
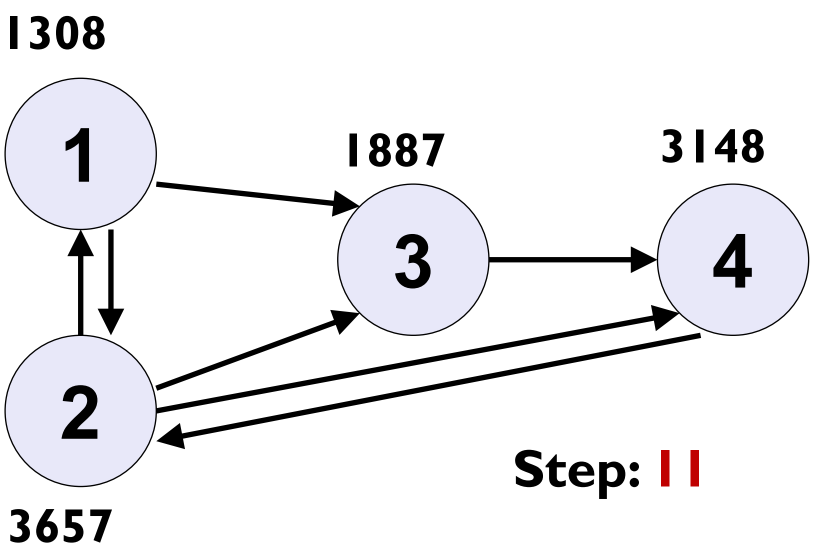
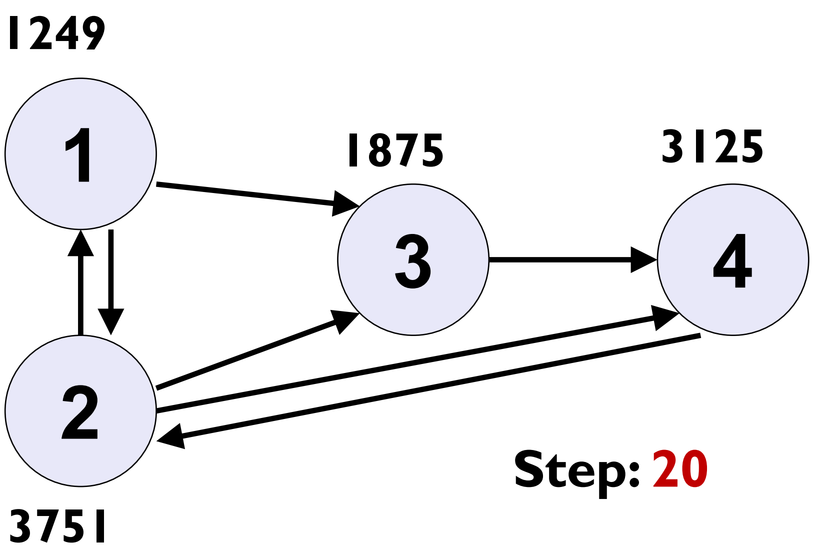


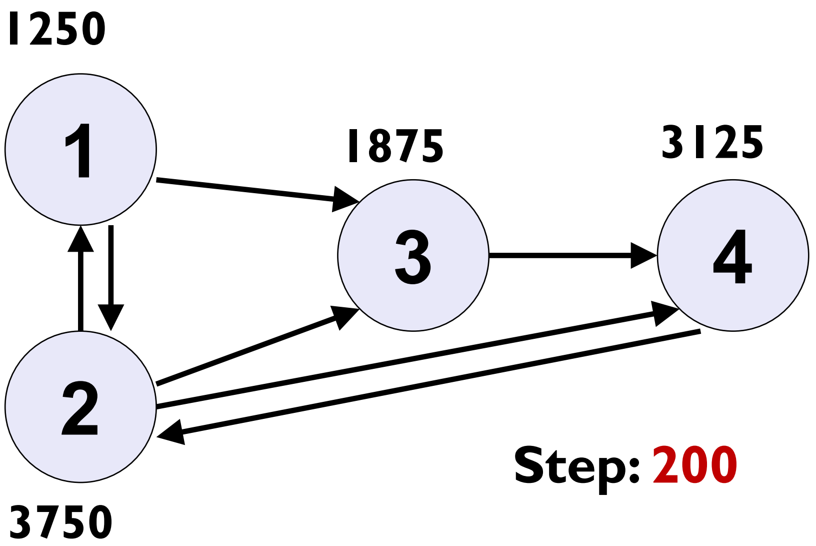
10.000 Random Surfers



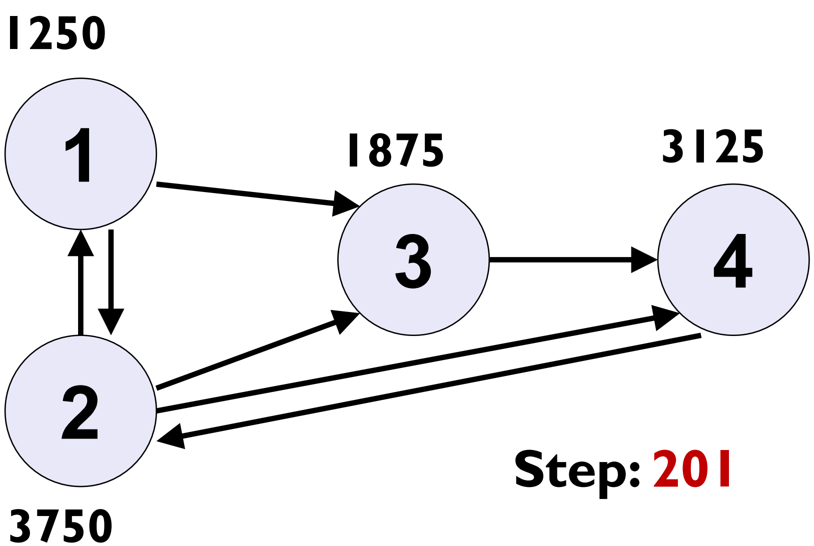
10.000 Random Surfers



10.000 Random Surfers



10.000 Random Surfers



Surfers visit some nodes more often than others.

These are those with many in-links from other frequently visited pages.

Pages visited more often has higher prestige (rank)

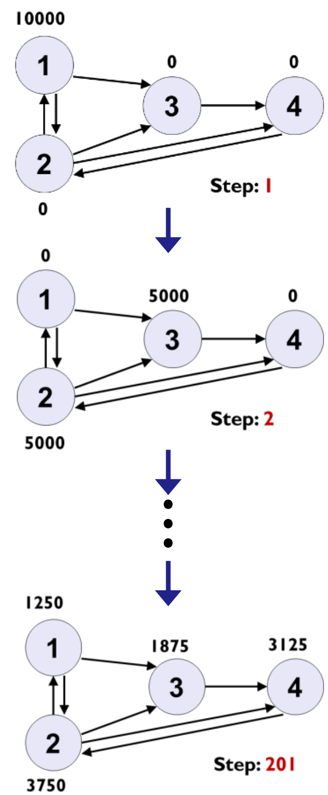
PageRank Intuition

Imagine users doing a random walk on web pages:

- Start on a **random** page
- At each step, continue to next page along one of the links on the current page, **equiprobably**

PageRank idea: Run to convergence; the rank (prestige) of a web-page is then proportional to:

- the **proportion** of random web-surfers that will be visiting the page at a given point in time
- = the **probability** that a random web-surfer is at this page at any point in time



$$(P(1), P(2), P(3), P(4)) = (0.1250, 0.3750, 0.1875, 0.3125)$$

PageRank – The Markov Chain Model
(Markov chains are abstractions of random walks)

The random surfer is described by

- An initial (step 0) probability distribution over all (n) web-pages

$$\boldsymbol{q}^{(0)} = \left(q_1^{(0)}, \dots, q_n^{(0)}\right)$$

- A transition probability matrix

$$\boldsymbol{P} = \begin{pmatrix} P_{1,1} & \cdots & P_{1,j} & \cdots & P_{1,n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ P_{i,1} & \cdots & P_{i,j} & \cdots & P_{i,n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ P_{n,1} & \cdots & P_{n,j} & \cdots & P_{n,n} \end{pmatrix}$$

Worth noticing:

- $P_{i,j} = 0$, if no link from i to j
  Happens many, many times!!!
- Each row sums to 1

$P_{i,j}$ = probability of moving from page ***i*** to page ***j***

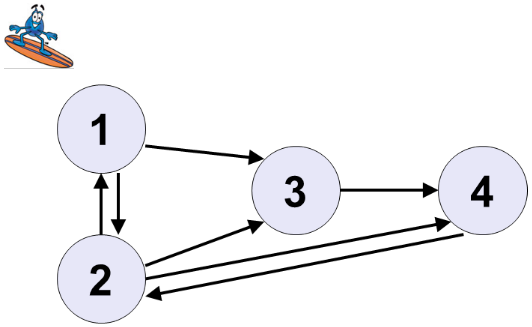
$$= \frac{1}{out\!-\!degree(i)}$$

Example (from before)

Initial probability distribution

(Here, surfer starts on page 1, but could have started anywhere!)

$$\boldsymbol{q}^{(0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}$$



Transition probability matrix

$$\boldsymbol{P} = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Markov Chain Transitions

$P_{i,j} = 0 \text{ when } i \not\to j$

The way we compute transitions:

$$q_j^{(t)} = \sum_{i=1}^{n} q_i^{(t-1)} P_{i,j} = \sum_{i:i\to j}^{n} q_i^{(t-1)} P_{i,j}$$

computed for all pages $j \in \{1, \dots n\}$

Matrix notation, the way we talk about it:

$$\boldsymbol{q}^{(t)} = \boldsymbol{q}^{(t-1)} \boldsymbol{P}$$

$$\left(q_1^{(t)} \dots q_j^{(t)} \dots q_n^{(t)}\right) = \left(q_1^{(t-1)} \dots q_j^{(t-1)} \dots q_n^{(t-1)}\right) \begin{pmatrix} P_{1,1} & \cdots & P_{1,j} & \cdots & P_{1,n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ P_{i,1} & \cdots & P_{i,j} & \cdots & P_{i,n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ P_{n,1} & \cdots & P_{n,j} & \cdots & P_{n,n} \end{pmatrix}$$

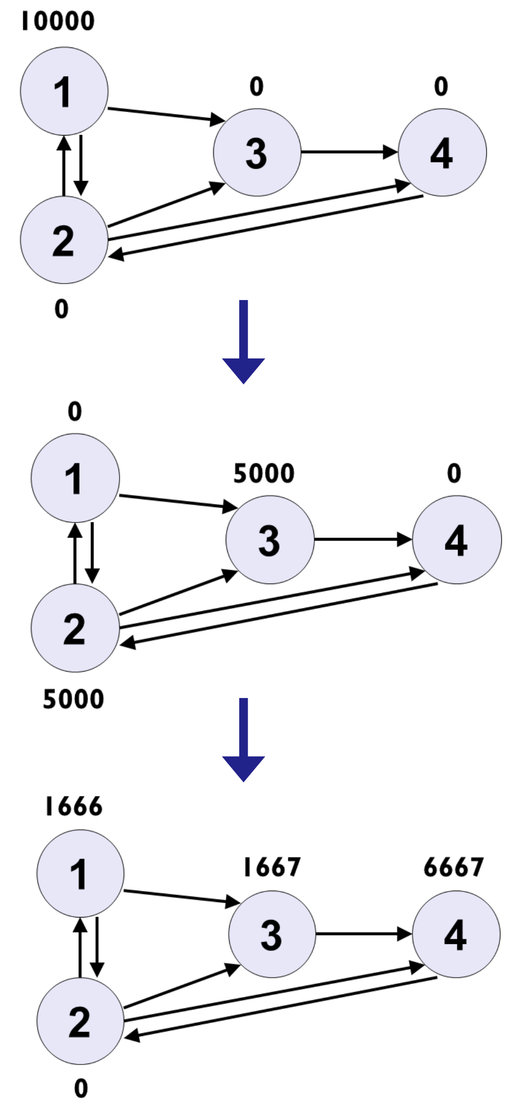
Example (cont.)

$$q^{(1)} = q^{(0)} P$$

$$\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$q^{(2)} = q^{(1)} P$$

$$\begin{pmatrix} \frac{1}{6} & 0 & \frac{1}{6} & \frac{4}{6} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$



…and for the Bayesian network folks!

PageRank is a (first order) Markov chain represented by a Bayesian network as follows

$$q^0 \xrightarrow{P} q^1 \xrightarrow{P} q^2 \xrightarrow{P} q^3 \xrightarrow{P} \ldots$$

PageRank – Stationary Distribution

$\boldsymbol{q}$ is stationary, if

$$\boldsymbol{q} = \boldsymbol{q}\boldsymbol{P}$$

Example

$$\begin{pmatrix} \frac{1}{8} & \frac{3}{8} & \frac{3}{16} & \frac{5}{16} \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & \frac{3}{8} & \frac{3}{16} & \frac{5}{16} \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Under *some conditions* (more detail on next slide)

- For any $\boldsymbol{q}^{(0)}$
- a Markov chain has a *unique* stationary distribution $\boldsymbol{q}^*$

$$\lim_{t\to\infty} \boldsymbol{q}^{(t)} = \boldsymbol{q}^*$$

"Eigen-Stuff"

$\boldsymbol{q}$ with $\boldsymbol{q} = \boldsymbol{q}\boldsymbol{P}$ is also called an Eigen-vector of $\boldsymbol{P}$ with Eigen-value 1.
In fact, $\boldsymbol{q}$ is the principal Eigen-vector, because $\boldsymbol{P}$ is a transision probalility matrix.

Many ways to find the principal Eigen-vector in practice:

- SVD
- Power iteration – PageRank algorithm

Stationarity conditions

The PageRank iterations $\boldsymbol{q}^{(t)} = \boldsymbol{q}^{(t-1)}\boldsymbol{P}$ must be representable by an **ergodic** Markov chain.

A Markov chain is ergodic if it is

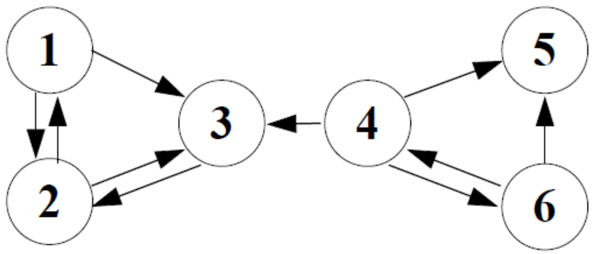
- *Irreducible*: “every node is reachable from every other node”
- *Aperiodic*: “nodes are not partitioned into sets such that all transitions occur cyclically from one set to another

Sufficient condition:

- A Markov chain is ergodic, if there is a strictly positive probability to pass from any state to any other state in one step.

We will be using this one

Complication: Dangling Pages



Ups! – A dangling page
$\Rightarrow$ ~~irreducible~~

"Transition matrix":

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix}$$

Problem: Not a proper transition matrix, because dangling pages (Ex.: page **5**) have no defined transitions

Q: What do we do now – our theory breaks down!

Teleporting

At a dangling node, jump to a random web page.

At any non-dangling node, with probability α (e.g.,10%), jump to a random web page.

- With remaining probability (1-α) (e.g., 90%), go out on a random link.
- α - a parameter. Should mirror our belief that a random web surfer would leave a page in another way than following a link

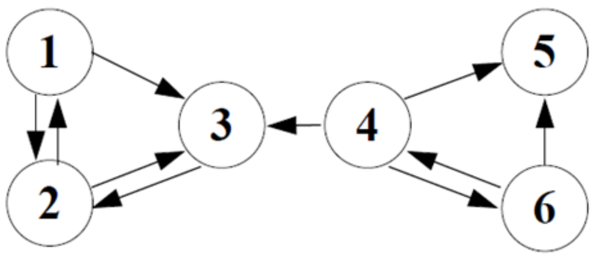
Result of teleporting:

- Now we cannot get stuck locally.
- there is a strictly positive probability to pass from any state to any other state in one step ⇒ guaranteed ergodicity
- The power iteration converges to a unique $\boldsymbol{p}^*$ ←PageRank

Teleporting …in math

New transition matrix: $\boldsymbol{P}_{PageRank} = (1-\alpha)\boldsymbol{P} + \alpha\boldsymbol{U}$

Example:



$$\boldsymbol{P}_{pageRank} = (1-\alpha)\begin{pmatrix} 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} + \alpha \begin{pmatrix} 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \end{pmatrix}$$

Definition of PageRank
(Brin and Page '98)

Consider the following infinite random walk (surf):

- Initially the surfer is at a random page
- At each step, the surfer proceeds
  - to a randomly chosen web page with probability α
  - to a randomly chosen successor of the current page with probability $1 - \alpha$

The PageRank of a page p is the fraction of steps the surfer spends at p in the limit. (A score between 0 and 1)

<u>Notice</u>: Using link structure only!

PageRank -- Summary

- The page rank of webpage *i* is

  $$n \cdot q_i^*$$

  where q_i^* is the (unique) limit distribution of the Markov chain defined by $\boldsymbol{P}_{PageRank}$

- We use Power Iterations to approximate the PageRank by iterating

  $$\boldsymbol{q}^{(t)} = \boldsymbol{q}^{(t-1)} \boldsymbol{P}_{PageRank}$$

  until $\boldsymbol{q}^{(t)}$ does not change very much

- Used as one of the ranking criteria in Google, Bing,…

SIMRANK: RANDOM WALK BASED SIMILARITY

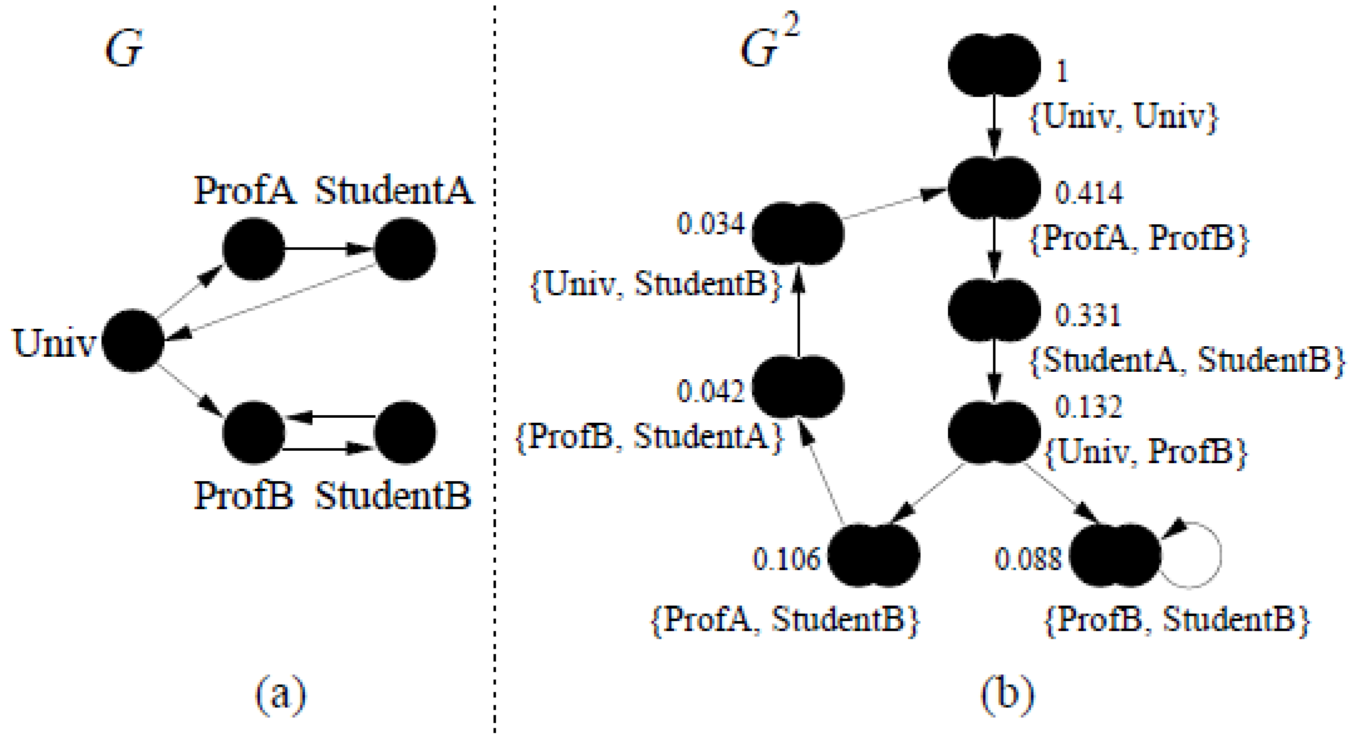
Structural Context

Similar objects are related to similar objects

Example:

Objects A and B are similar if they are related to objects C and D respectively and C and D are themselves similar

Example


G
ProfA StudentA
Univ
ProfB StudentB
(a)
G^2
1
{Univ, Univ}
0.414
{ProfA, ProfB}
0.034
{Univ, StudentB}
0.331
{StudentA, StudentB}
0.042
{ProfB, StudentA}
0.132
{Univ, ProfB}
0.106
{ProfA, StudentB}
0.088
{ProfB, StudentB}
(b)

Note

Observe, that for different applications such graph may be constructed differently, may exist explicitly, or may need to be formed from the data as we had in other lectures.

Typical relations:

- co-citations
- Co-purchases
- Reference
- …

Model

$$G = (V, E)$$

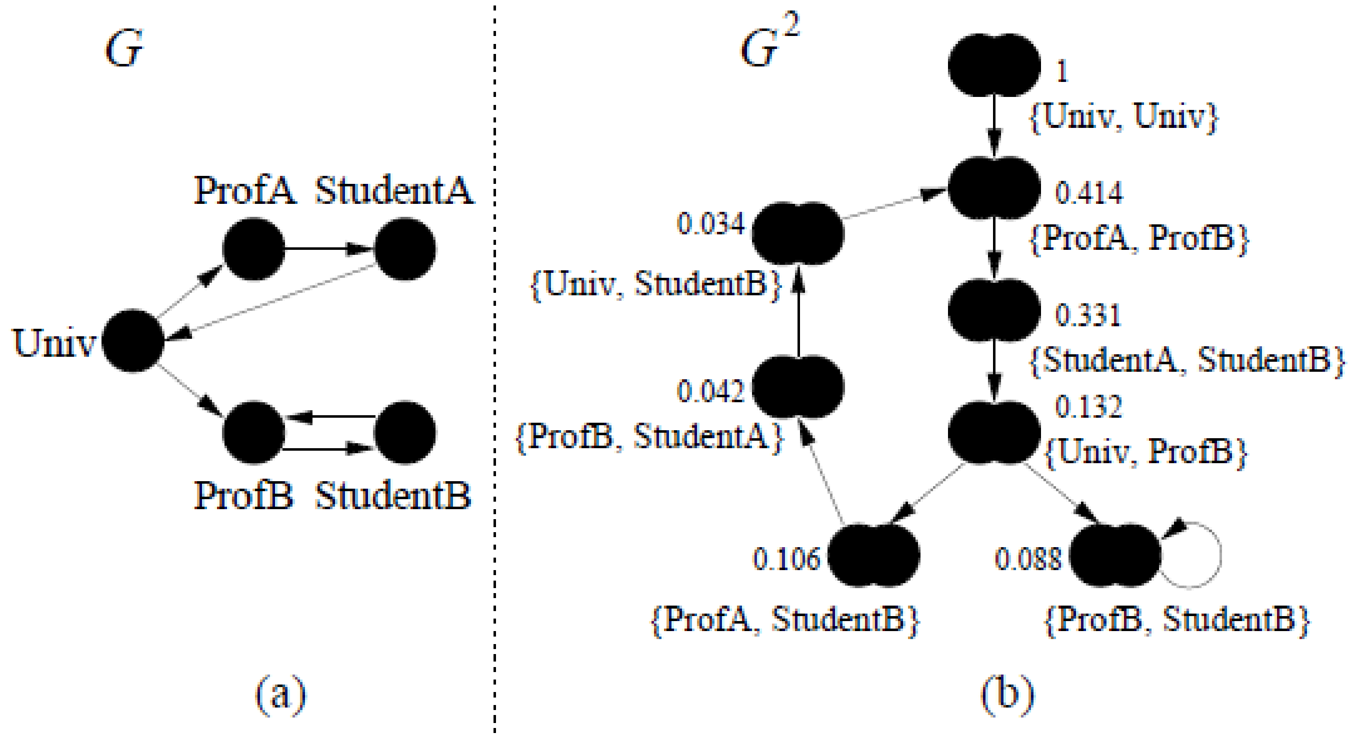
V… objects
E … relationships between objects
$I(v)$… Set of in neighbors of v
$O(v)$… Set of out neighbors of v
$I_i(v)$… individual in neighbor of v
$O_i(v)$… individual out neighbor of v

Homogeneous domain: nodes usually represent documents and relations are usually hyperlinks or reference

User – Item domain: domain is bipartite, users and items are nodes, and a relation represents usually a purchase or an expression of preference

Example

Simplified, symmetric pairs are ommitted



Similarity

Object is maximally similar to itself with the score 1

Such similarity can be predefined for other objects as well but we will not deal with them here

From prev. figure: Prof A and Prof B are similar because they are referenced by node Univ which is similar to itself

Student A and Student B are similar because they are referenced by similar Prof A and Prof B

Similarity

$$s(a,b) = \begin{cases} 1, if\ a = b \\ 0, if\ I(a) = \emptyset\ or\ I(b) = \emptyset \\ \frac{C}{|I(a)||I(b)|} \sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} s(I_i(a), I_j(b)) \end{cases}$$

C… is a constant between 0 and 1

This formula is written for every pair of nodes in the graph leading to n^2 formulaes

Model for propagation of similarity

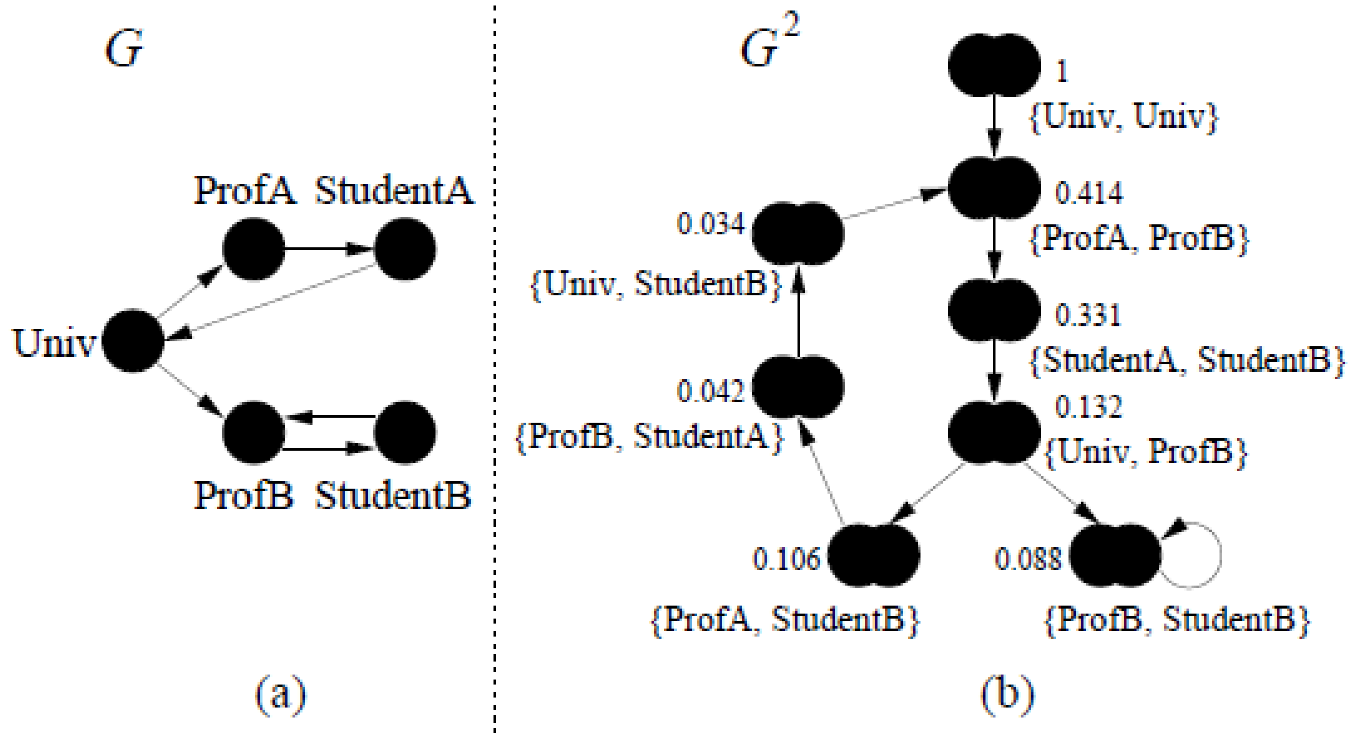
$$G^2 = (V^2, E^2)$$

$V^2 = V \times V$ represents a pair (a, b) of nodes in the graph G

Edge from (a, b) to (c, d) in E^2exists iff edge (a, c) and (b, d) exist in E (or G)

Example

Simplified,
symmetric pairs are omitted
Pairs with 0 similarity are ommitted



Constant C

Steers the confidence level or decay factor

Example:

s(x, x) = 1 and x is related to c and d.

We do not want to say s(x, x) = s(c, d) = 1

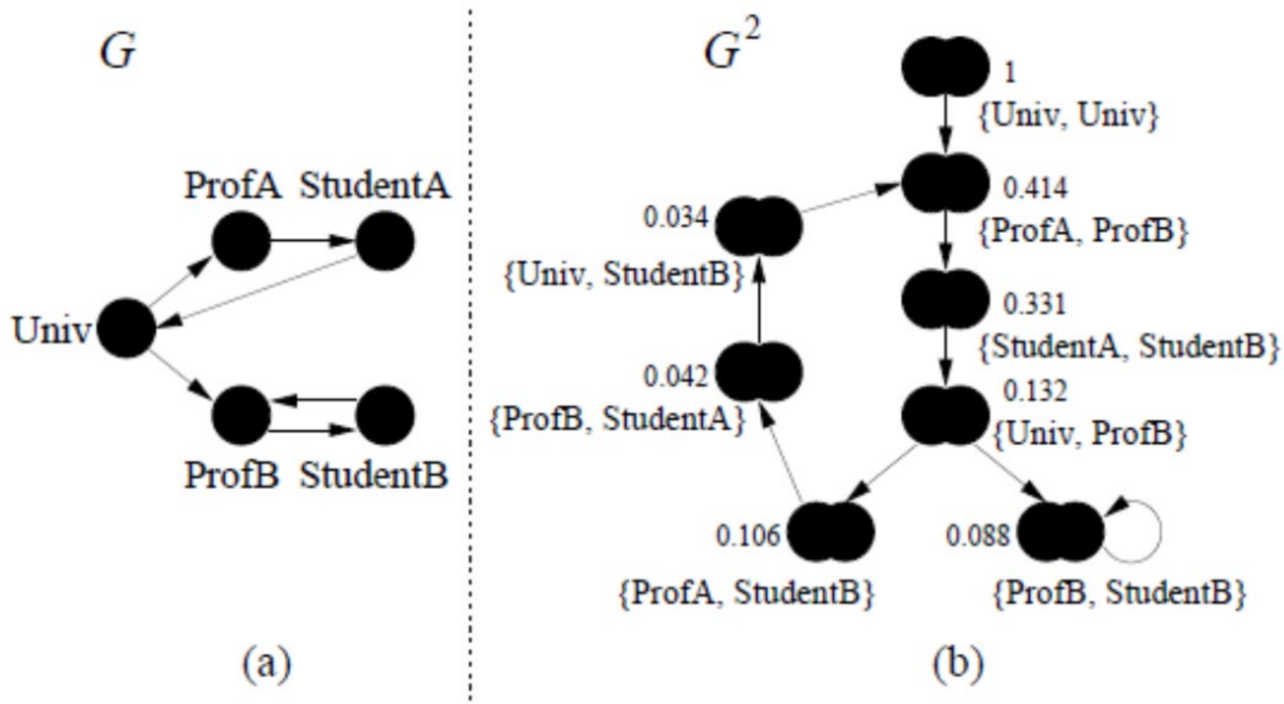
We want to say that s(c, d) = C*s(x, x)

We are less confident in s(c, d)

We can also see it as a rate of decay as the similarity flows across edges (C < 1)

In the paper C = 0,8

Bipartite SimRank in homogeneous domain



Example (from paper)

Students take group of similar courses
Student A of CS can take electives from sociology related group
Student B of CS can take electives English related group

Instead of comparing all course of student A and student B we can compare only elective
Or in minmax case on one course the most similar

Interation to the fixed point – homogeneous domain

$$R_0(a,b) = \begin{cases} 0, (if\ a \neq b) \\ 1, (if\ a = b) \end{cases}$$

$$R_{k+1}(a,b) = \begin{cases} 1, (if\ a = b) \\ \dfrac{C}{|I(a)||I(b)|} \displaystyle\sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} R_k(I_i(a), I_j(b)), (if\ a \neq b) \end{cases}$$

Interation to the fixed point – homogeneous domain

Initialization

$$R_0(a,b) = \begin{cases} 0, (if\ a \neq b) \\ 1, (if\ a = b) \end{cases}$$

$$R_{k+1}(a,b) = \begin{cases} 1, (if\ a = b) \\ \dfrac{C}{|I(a)||I(b)|} \sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} R_k(I_i(a), I_j(b)), (if\ a \neq b) \end{cases}$$

Update/Iteration

Notes

Values R_k are non decreasing with k increasing
Relative rankings stabilize soon ($K \approx 5$)
Paper shows that the algorithm converges to the real s(a, b) scores in limit

Link Prediction Realization

- Define a **similarity measure S(v;w)** between nodes based on graph structure
- Use **S(v;w)** directly as a **ranking** measure, i.e. for each node, rank not connected nodes according to the similarity, which gives for each node a rank
- Use a testing measure for ranked lists
- The goal of course is to have the intended connections as high as possible, and therefore measures which can consider position in the ranked list better reflect the situation
- In the self study, I suggested HIT rate which does not consider the position on the rank
- Feel free to use other such as NDCG or MRR.

- HIT = | Link Prediction Lists where a Node Appeared | / |All Link Prediction Lists|
- MRR = 1/ | All Link Prediction Lists | * sum_over_lists(1 / rank(i))
    - rank(i) is a position within particular list

Conclusions

Similarity based link prediction can be based on local and global measures

Already simple local measures work well

Global measures are more difficult to spam